

Developer Mock Test Results

Student: Shiva kasula

Name:	Shiva kasula	Student ID:	112
Date:	2026-03-05 12:31:46	Performance:	Good
Overall Score:	16.4/25 (65.6%)	Test ID:	b24ded1c-385b-4847...
Warnings:	0/3	Status:	Completed

Section-wise Performance

Section	Score	Percentage	Status
Aptitude	5/10	50.0%	Pass
MCQ/Theory	9/10	90.0%	Pass
Coding	2.4/5	48.0%	Needs Work

Detailed Question Review

APTITUDE SECTION (5/10 - 50.0%)

Q1. If it is true that all A are B and some B are C, what can be concluded about A and C?

Your Answer: Some A are C

Correct! Well done.

Q2. A mixture of paint and water is in the ratio 2:3. If 2 cups of paint are added to the mixture, what is the new ratio of paint to water?

Your Answer: 4:3

Excellent! Right answer.

Q3. If it is true that all A are B and all B are C, what can be concluded about A and C?

Your Answer: Some A are C

Correct Answer: All A are C

The correct answer is "All A are C" because if all A are B and all B are C, then it logically follows that all A are also C. The user's mistake is likely due to not fully applying the transitive property of the given statements, which allows us to conclude a direct relationship between A and C. By not recognizing this chain of relationships, the user incorrectly settled on "Some A are C", which does not fully capture the certainty of the relationship between A and C.

Q4. A product is on sale for 25% off its original price of \$120. How much will you pay for the product?

Your Answer: No answer (Skipped)

Correct Answer: \$90

■ To find the sale price, calculate 25% of the original price of \$120 and subtract it from \$120. The key calculation step is: $\$120 * 0.25 = \30 , then $\$120 - \$30 = \$90$. Therefore, the correct answer is \$90, which matches the given correct answer, and this is the amount you will pay for the product.

■ **Q5. If 18 workers can build a house in 12 days, how many workers will be needed to build the same house in 10 days?**

Your Answer: 22 workers

■ Correct! Good understanding.

■ **Q6. A store buys a product for \$40 and sells it for \$35. What is the loss percentage?**

Your Answer: No answer (Skipped)

Correct Answer: 12.5%

■ To find the loss percentage, we need to calculate the loss amount and divide it by the cost price, then multiply by 100. The key calculation step is: $\text{Loss percentage} = ((\text{Cost price} - \text{Selling price}) / \text{Cost price}) * 100 = ((40 - 35) / 40) * 100 = (5 / 40) * 100 = 12.5\%$. The correct answer is indeed 12.5%, which means the store incurred a loss of 12.5% on the product.

■ **Q7. A shirt is on sale for 15% off its original price of \$80. How much will you pay for the shirt?**

Your Answer: \$65

Correct Answer: \$68

■ To find the sale price, calculate 15% of the original price of \$80: $0.15 * \$80 = \12 . Then, subtract the discount from the original price: $\$80 - \$12 = \$68$. The user's answer of \$65 is likely incorrect because they may have miscalculated the discount amount or subtracted it incorrectly from the original price.

■ **Q8. A product is on sale for 20% off its original price. If the original price is \$100, how much will you pay for the product after the discount?**

Your Answer: \$80

■ Correct! Well done.

■ **Q9. The average of 8 numbers is 12. If one of the numbers is 15, what is the sum of the remaining 7 numbers?**

Your Answer: 80

■ Correct! Well done.

■ **Q10. A mixture of juice and water is in the ratio 3:5. If 3 cups of juice are added to the mixture, what is the new ratio of juice to water?**

Your Answer: 3:5

Correct Answer: 6:5

■ The correct answer is 6:5, which can be found by first assuming an initial amount of juice and water, such as $3x$ and $5x$, where x is a multiplier. If 3 cups of juice are added, the new amount of juice is $3x + 3$, and since the ratio of juice to water is given as 3:5, we can set up a proportion to find the new ratio: $(3x + 3) / 5x = 6 / 5$ when $x = 1$, showing the new ratio is indeed 6:5. The user's mistake was likely assuming the ratio remains the same after adding juice, without recalculating the new ratio based on the added amount.

■ MCQ SECTION (9/10 - 90.0%)

■ **Q1. What type of software is used to write and edit Java code?**

Your Answer: Compiler

Correct Answer: Code Editor or IDE

■ The correct answer is B) Code Editor or IDE because it provides a comprehensive environment for writing, editing, and managing Java code, offering features such as syntax highlighting, code completion, and project management. The user's answer, Compiler, is incorrect because a compiler is a software that translates Java code into machine code, but it is not typically used for writing or editing code. The user likely mistook the role of a compiler, which is a crucial step in the development process, but not the primary tool for authoring Java code.

■ Q2. What is inheritance in Java?

Your Answer:

```
A subclass inherits properties and behavior from a parent class
```

■ Correct! Well done.

■ Q3. What should be followed in Java?

Your Answer: Java coding standards

■ Excellent! Right answer.

■ Q4. When should comments be written in Java?

Your Answer: For complex logic

■ Correct! Good understanding.

■ Q5. What is the process of converting one data type to another in Java?

Your Answer: Casting

■ Excellent! Right answer.

■ Q6. How does Java provide error handling?

Your Answer: Using try, catch, and finally blocks

■ Well done!

■ Q7. What is optimization in Java?

Your Answer: The process of improving the performance of a program

■ Excellent! Right answer.

■ Q8. How does Java handle errors?

Your Answer: Using try, catch, and finally blocks

■ Well done!

■ Q9. What is exception handling in Java?

Your Answer: The process of handling runtime errors

■ Well done!

■ Q10. What are classes in Java?

Your Answer:

```
Blueprints for objects
```

■ Correct! Well done.

■ CODING SECTION (2.4/5 - 48.0%)

■ Q1. Write a Java program that reads an integer from stdin and prints the Fibonacci series up to that number. ****Input Format:**** - Line 1: An integer ****Output:****

Your Answer:

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        // Write your solution here
        solution();
    }
}
```

```

    }

    public static void solution() {
        // Your code here
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();

        int a = 0, b = 1;

        while (a <= n) {
            System.out.print(a + " ");
            int next = a + b;
            a = b;
            b = next;
        }

        sc.close();
    }
}

```

■ Your code has successfully passed all test cases, demonstrating a thorough understanding of the problem and its solution. Great job on achieving an "Accepted" status, indicating that your code is correct and effective.

■ **Q2. Write a Java program that reads a string from stdin and checks if it is a valid time in the format "HH:MM:SS". **Input Format:** - Line 1: A string ...**

Your Answer:

```

import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        // Write your solution here
        solution();
    }

    public static void solution() {
        // Your code here
        Scanner sc = new Scanner(System.in);
        String time = sc.nextLine();

        String[] parts = time.split(":");

        if (parts.length != 3) {
            System.out.println("Not Valid Time");
            return;
        }

        try {
            int hh = Integer.parseInt(parts[0]);
            int mm = Integer.parseInt(parts[1]);
            int ss = Integer.parseInt(parts[2]);

            if (hh >= 0 && hh <= 23 && mm >= 0 && mm <= 59 && ss >= 0 && ss <= 59) {
                System.out.println("Valid Time");
            } else {
                System.out.println("Not Valid Time");
            }
        } catch (Exception e) {
            System.out.println("Not Valid Time");
        }
    }
}

```

```

        sc.close();
    }
}

```

■ Your code has successfully passed all test cases, demonstrating a thorough understanding of the problem and its requirements. Great job on achieving a perfect score and receiving an "Accepted" status!

■ **Q3. Write a Java program that reads a string from stdin and checks if it is a valid email address. **Input Format:** - Line 1: A string **Output Format:...**

Your Answer:

```

import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        // Write your solution here
        solution();
    }

    public static void solution() {
        // Your code here
        Scanner sc = new Scanner(System.in);
        String email = sc.nextLine();

        String pattern = "[A-Za-z0-9+_.-]+@[A-Za-z0-9.-]+$";

        if (email.matches(pattern)) {
            System.out.println("Valid Email");
        } else {
            System.out.println("Not Valid Email");
        }

        sc.close();
    }
}

```

Correct Answer:

```

import java.util.Scanner;
import java.util.regex.Pattern;

public class Main {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        String email = scanner.nextLine();
        String regex = "[A-Z0-9._%+-]+@[A-Z0-9.-]+\\.[A-Z]{2,6}$";
        if (Pattern.compile(regex, Pattern.CASE_INSENSITIVE).matcher(email).find()) {
            System.out.println("Valid Email");
        } else {
            System.out.println("Not Valid Email");
        }
        scanner.close();
    }
}

```

■ The student's code failed 3 out of 5 test cases with a Runtime Error, indicating that the issue is likely due to an unhandled exception or an infinite loop. The provided code snippet does not reveal any obvious issues, suggesting that the problem may lie in the unseen parts of the code. The "Runtime Error" message implies that the code compiles but encounters an error during execution, which may be caused by the way the input is handled or the regular expression pattern used to validate the email address.

■ **Q4. Write a program that reads a string and prints the string with all vowels removed. **Input Format:** - Line 1: Input string (string) **Output Format:...**

Your Answer:

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        // Write your solution here
        solution();
    }

    public static void solution() {
        // Your code here
        Scanner sc = new Scanner(System.in);
        String email = sc.nextLine();

        String pattern = "[A-Za-z0-9+_.-]+@[A-Za-z0-9.-]+$";

        if (email.matches(pattern)) {
            System.out.println("Valid Email");
        } else {
            System.out.println("Not Valid Email");
        }

        sc.close();
    }
}
```

Correct Answer:

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        String input = scanner.nextLine();
        String vowels = "aeiouAEIOU";
        StringBuilder result = new StringBuilder();

        for (char c : input.toCharArray()) {
            if (vowels.indexOf(c) == -1) {
                result.append(c);
            }
        }

        System.out.println(result.toString());
        scanner.close();
    }
}
```

■ The student's code is checking if the input string is a valid email, but the problem requires removing vowels from the input string. As a result, the code is printing "Not Valid Email" instead of the expected output "Hll Wrld". The code is not implementing the required functionality of removing vowels from the input string.

■ **Q5. Write a Java program that defines a Shape class with methods to calculate area and perimeter, and a Circle class that inherits from Shape. **Input Fo...**

Your Answer:

No answer (Skipped)

Correct Answer:

```
import java.util.Scanner;

class Shape {
    protected double radius;
```

```

    public Shape(double radius) {
        this.radius = radius;
    }

    public double calculateArea() {
        return Math.PI * radius * radius;
    }

    public double calculatePerimeter() {
        return 2 * Math.PI * radius;
    }
}

class Circle extends Shape {
    public Circle(double radius) {
        super(radius);
    }
}

public class Main {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        int numShapes = scanner.nextInt();
        scanner.nextLine(); // Consume newline left-over

        for (int i = 0; i < numShapes; i++) {
            String shapeType = scanner.nextLine();
            double radius = scanner.nextDouble();
            scanner.nextLine(); // Consume newline left-over

            if (shapeType.equals("Circle")) {
                Circle circle = new Circle(radius);
                System.out.println(String.format("%.1f %.1f", circle.calculateArea(),
circle.calculatePerimeter()));
            } else if (shapeType.equals("Shape")) {
                Shape shape = new Shape(radius);
                System.out.println(String.format("%.1f %.1f", shape.calculateArea(),
shape.calculatePerimeter()));
            }
        }

        scanner.close();
    }
}

```

■ There are no test case results provided to analyze the failure. The student code is also not submitted, making it impossible to determine the cause of the failure. Without the actual test case results or the student's code, it's not possible to provide accurate feedback.

■ Recommendations

Areas that need improvement:

- **Coding:** Practice more coding problems on platforms like LeetCode or HackerRank.